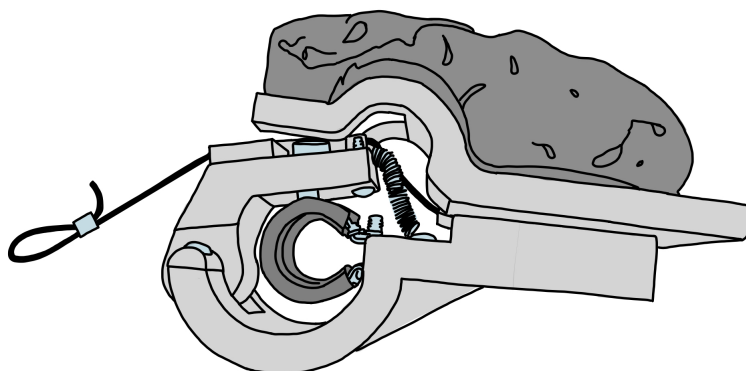


VeloBrake Assist

An Assistive Braking Device for Biking

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Open Access Report

Assistive Technology Challenge 2026

Group: P6, Adapted environment for ectrodactily

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1. Introduction

VeloBrake Assist is an assistive device developed during a semester-long interdisciplinary project called Assistive Technology Challenge, organized by HackaHealth at EPFL (École Polytechnique Fédérale de Lausanne). Each team, composed of 4 Master's students, was assigned a "challenger" with a specific need to solve an everyday challenge, adopting a "user-in-the-loop" approach.

The challenger assigned to this team is affected by ectrodactyly, also called split hand/foot malformation (SHFM), a rare congenital condition in which the development of the hand or foot is disturbed, causing the deficiency or absence of one or more central digits. The hands and feet of people with ectrodactyly are often described as "claw-like" due to their V-shaped appearance [3]. The challenger presented with one split hand, the left one, with the right one completely functional.

During the first meeting, the challenger described a daily life filled with constraints and limitations, and the continuous feeling of living in a world not shaped for them as they have issues with holding and grasping objects. One of the main things they wished to enjoy is biking. However, the conformation of their left hand does not allow them to grip the handlebar securely or to brake. Their goal was for the team to develop a device that could allow them to bike safely.

In this report, the development of VeloBrake Assist will be discussed, detailing the design choices and the iterative prototyping process behind them. Further steps needed to fully satisfy the needs of the challenger will also be presented.

2. Design Choices

2.1. User Requirements

2.1.1 Safety

As mentioned in the Introduction (Section 1), the user's primary requirement is safety ("must-have"). To maneuver the bike safely, the user needs to grip the handlebars securely and to brake with both hands, not only the functional one. The braking mechanism should be modifiable, allowing the user to slow down gradually rather than only stop abruptly, and should not activate accidentally. Additionally, immediate release of the handlebar must be possible in case of emergency, such as a fall. This was a critical limitation of a previous assistive tool they used (Figure 6.H) and the main reason they never felt fully safe while biking. Lastly, it should be sturdy enough to withstand the forces experienced while biking and braking.

2.1.2 Comfort

Comfort is also a key requirement ("must-have"); the device should therefore be mounted on the bike rather than worn. It should not feel bulky, but shaped to fit the user's hand, and it should allow for natural and coordinated control of the bike.

2.1.3 Portability

Additionally, portability was identified as a desirable feature ("nice-to-have"). The device should be easily mountable, removable, and compatible with different bikes. The ideal scenario described by the user would be to carry the device in a backpack and mount it on any city bike when needed.

2.1.4 Support for Chronic Back Pain Management

Lastly, the user also reported suffering from chronic back pain, resulting from the overuse of the right hand to perform most daily actions. This leads to muscular imbalance between the two sides of the body, causing uneven load on the spine and additional issues while biking. While this condition is not directly addressed by the device, the user expressed the need for a solution that could help reduce this imbalance over time.

2.2. System Requirements

The user's needs were translated into system requirements that guided the design of the device.

2.2.1 Braking System

The core of the device will be a braking system based on direct mechanical actuation of the brake through a cable. Braking shall be activated by a downward push of the wrist, a natural and comfortable movement for the user, and it shall be proportional to the force applied. The mechanism should be entirely passive, with no electronic components, ensuring simplicity and reliability. A spring return mechanism will be integrated to prevent accidental activation.

2.2.2 Hand-grip

To ensure both comfort and a secure grip, the device should incorporate a hand grip shaped to fit the user's hand specifically, made from a soft material to maximize comfort during use.

2.2.3 Cable & Attachment Mechanism

Portability and compatibility will be addressed through two main design choices. First, a cable-based actuation system to ensure compatibility with various brake designs. Second, a secure attachment mechanism to fit the standardized handlebar diameter (removing the handlebar's coatings), allowing the device to be mounted and removed from different bikes.

2.2.4 Weighted Sleeve

Finally, a weighted forearm sleeve should be used to address the muscular imbalance caused by chronic overuse of the right hand. A sleeve design should be chosen over conventional weighted tools, as the latter concentrate load on a single point of the arm, whereas the sleeve distributes it evenly across the forearm, which ensure comfortability for the user.

2.3. Final Design Specifications

Write about materials (PET-CF for robustness), specific design choices (eg the hole to fasten the attachment mechanism's screw, type of screws), key design parameters.. The final design was build specifically for the challenger's bike which was used for validation and verification, prioritizing robustness and safety on the user's bike first and then compromising with some design specifications in order to create something that could be adaptable to different bikes.

2.3.1 Braking Mechanism: 3D designed parts, cable, spring

2.3.2 Hand - grip

I think we also need to ask sebastian which final parameters where used for the printing of TPU.

2.3.3 Clamping Collar & Fastening Screw

2.3.4 LaceUp Arm Sleeves

After discussing with the user, the "LaceUp Arm Sleeves" (Figure 6.J), a pair of weighted compression sleeves made with LYCRA® fiber weighing 1 lb each, were purchased from LaceUp (size Small/Medium) [1], an American brand specializing in weighted sport garments.

3. Reproducing Prototype

This is where the iterative, hands-on reality lives. Because design and prototyping were coupled for you, structure this as rounds of iteration rather than a linear list of steps:

3.1. 3D Printing Iterations

3D printing iterations (what changed between versions and why)

3.2. Molded Hand - grip

Obtaining an accurate mold of the user's hand was the most user-specific step of the entire process. During one of the early sessions with the user, Pâte Fimo (a polymer clay) was used to capture the natural grip position of the left hand directly on the handlebar of the user's own bike using a platform in PETG similar to the one used in the final design. This approach was key: it ensured that the resulting shape reflected not an idealized grasp, but the actual posture adopted during riding. The hardened Fimo mold was then 3D scanned to obtain a digital model, which was cleaned and prepared for printing. The first print was made in PETG to verify the fit and general shape. While dimensionally accurate, the rigidity of the material made the grip uncomfortable against the user's skin. A second version was printed in TPU (thermoplastic polyurethane) to introduce softness. After user testing, the printing parameters were adjusted reducing infill and perimeter to obtain a softer and more compliant result that the user found comfortable and secure.

3.3. Attachment Mechanism

Attachment mechanism trials (what failed, what worked)

3.4. Bike Repair

The user's bicycle required significant maintenance. The bicycle was collected from the user's home and transported to PointVelo, the on-campus bicycle repair service, where the necessary repairs were carried out with the help of the staff.

The kickstand was first re-secured, as it had become loose due to some missing fastening screws. The tires were then inspected, as the user had reported rapid pressure loss. No punctures were found and the issue was attributed to the prolonged inactivity of the bike. The tires were then reinflated.

The brake cables were adjusted to reduce lever travel distance. This was achieved by partially unscrewing the cable anchor bolt on each brake caliper, pulling the cable through to take up the slack, and re-securing it. This adjustment was necessary to shorten the actuation range of the levers and free up sufficient clearance on the handlebar for the device to be mounted.

Finally, recurrent chain derailment during gear shifting was addressed by installing a rubber-tipped set screw on the seat tube, acting as a mechanical stop to prevent the chain from dropping off during gear changes.

3.5. User testing sessions for Validation and Verification

their feedback, and what it changed

4. Further Improvements

The current prototype successfully demonstrates the core concept of VeloBrake Assist, but several limitations were identified during the project that would need to be addressed before the device could be considered production-ready.

4.1. Metal Shaft

The most pressing mechanical concern is the long-term durability of the 3D-printed parts. Although PET-CF was chosen for its stiffness, the pivot axis of the lever is a point of concentrated stress and would benefit from being replaced by a metal shaft machined to tolerance. This upgrade would significantly increase the lifespan of the device and its resistance to repeated braking cycles.

4.2. Cable Clamp

The cable attachment is another area for improvement. Currently, the cable is terminated with a crimped end which, while functional, requires a specific tool to adjust or replace. A cable clamp system was designed as a next step to allow tool-free cable length adjustment, making the device more accessible for the user to install and transfer between different bikes. However, the 2mm-cable clamp that was purchased [2] proved too bulky, obstructing the sliding of the cable and limiting the braking action, and was therefore not implemented in the final prototype.

4.3. Stability Across Different Bikes

The current attachment mechanism was designed around the user's own bike, with the handlebar grip removed to ensure a secure fit. While this approach achieves good stability on the current setup, transferring the device to a different bike may introduce variability due to differences in handlebar and brake geometry.

A promising avenue to address this would be a dual-point attachment system. A primary, rigid fixation point just inboard of the grip (where handlebar diameters are largely standardised across city bikes) would bear the majority of the braking forces. A secondary, lighter contact point at the grip itself would provide rotational stability without requiring a tight clamp, accommodating slight variations in grip diameter found across different bikes. This configuration could preserve the stability achieved on the current prototype while making the device genuinely transferable. However, the feasibility of this approach depends critically on the clamping force achievable at the primary fixation point: sufficient torque and a secure fastening system would be essential to ensure the device does not rotate or slip under braking load.

4.4. Mechanical and Environmental Testing

The prototype was validated through user testing sessions but has not undergone formal mechanical testing. Quantifying the forces involved during braking at the pivot axis, the cable attachment point, and the handlebar clamp would allow the design to be optimised and failure modes to be identified. This is particularly relevant if the device were to be reproduced or adapted for other users, as the current prototype should be regarded as a proof of concept.

Additionally, the device has not been tested under real-world weather conditions. Rain, temperature variation, and UV exposure are all likely to affect the mechanical properties of the printed parts and the cable over time. Weatherproofing the design through appropriate material selection and protective coatings would be necessary for reliable everyday outdoor use.

4.5. Two-hands device

Early in the design process, the possibility of developing a two-handed version of the device was considered, motivated by the user's requirement for natural and coordinated bilateral movement. Such a configuration would place an assistive device on both handlebars, allowing the user to brake with a coordinated movement.

However, this concept was not pursued for two main reasons. First, the right handlebar controls gear shifting; attaching a fixed grip platform on this side would make it difficult

for the user from operating the gear lever. Second, having both hands resting on the device rather than one directly on the handlebar poses a significant safety risk in case of mechanical failure. Given that the prototype has not undergone formal mechanical testing, it was considered essential that the user retains at least one hand in direct contact with the handlebar at all times, as a safety measure. The priority was therefore to validate the braking function on the left side first, leaving the two-handed configuration as a potential avenue for future exploration once the device has been thoroughly tested.

5. Conclusions

6. Figures



Figure 1: A. B. C. D. E. F. G. H. Frontal and dorsal view of the tool that the user previously used for biking; I. J. "LaceUp Arm Sleeves" purchased for the management of the challenger's chronic back pain.

References

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